Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Valtechies**
b. Team leader name: **Tejas Bhorade**
c. Problem Statement: **Sustainability & Social Impact — fuel and CO2e for farm-machinery fleets, with honest confidence**

Brief about the idea

FieldCarbon: where a farm-machinery fleet wastes fuel, what to change, and how sure we are. Every figure traceable.

The problem

- Agricultural contractors and co-operatives run tractors across many client farms.
- Buyers now ask them for fuel and CO2e **per farm and per job**.
- Fuel is bought in bulk, dispensed from one depot tank and burned across dozens of clients; the paperwork is on paper or in people's heads.
- Carbon tools give one number with **no error bar** and no next step.

FieldCarbon

- Turns telemetry, fuel logs, **photos of delivery notes and job sheets**, and **voice notes** into fuel, cost and CO2e by site, machine, job, field and client.
- Finds waste and tests its causes; ranks what to change by **strength of evidence**.
- Proves savings before and after; writes a **traceable statement per client farm**.

Gemini reads and explains. Tested code does every calculation. An auditor checks every number Gemini writes.

Opportunities

How is it different?

- Other tools give **a number**. FieldCarbon says **what to change, how sure we are, and what data would make us surer**.
- Every figure carries a label as text: **MEASURED, ALLOCATED or ESTIMATED**, and opens into its sources, emission factor and calculation.
- Savings are **ranges** from published research, never a single promise.

How does it solve the problem?

- Uses data contractors already have: telemetry CSVs, a fuel log, **photos of paperwork**, voice notes.
- Splits fleet fuel to **each client farm and job**, so a contractor can answer a buyer's request.
- Checks deliveries vs dispenses vs fuel burned, and **tests the causes** of each gap.
- Ranks only changes that fit this fleet.

USP

- **Uncertainty is the product:** labels on 100% of figures, and "what would improve this answer?" names the one missing input that most narrows a range.
- **Gemini never does the maths:** an auditor traces every number in an AI answer to a tool output.
- **Proof, not promises:** savings judged against machines that did not change.

List of features offered by the solution

Carbon ledger
Tonnes emitted, a bounded range that could be avoided, tonnes proven avoided, goal check.

Week summary
The three things to look at first, chosen by fixed rules in code.

Sites, machines, jobs
The same job compared across machines: L/ha, L/h, idle share.

Client farms
Fuel and CO2e for every client farm the fleet works for.

Fuel control
Depot deliveries vs dispenses vs fuel burned, month by month.

Delivery-note reading
Gemini reads photos (also live upload); code checks them against the fuel log.

Anomaly diagnosis
Code tests a fixed menu of causes; Gemini suggests; a person decides.

Voice notes & job sheets
Gemini hears machine faults and reads handwritten workshop sheets.

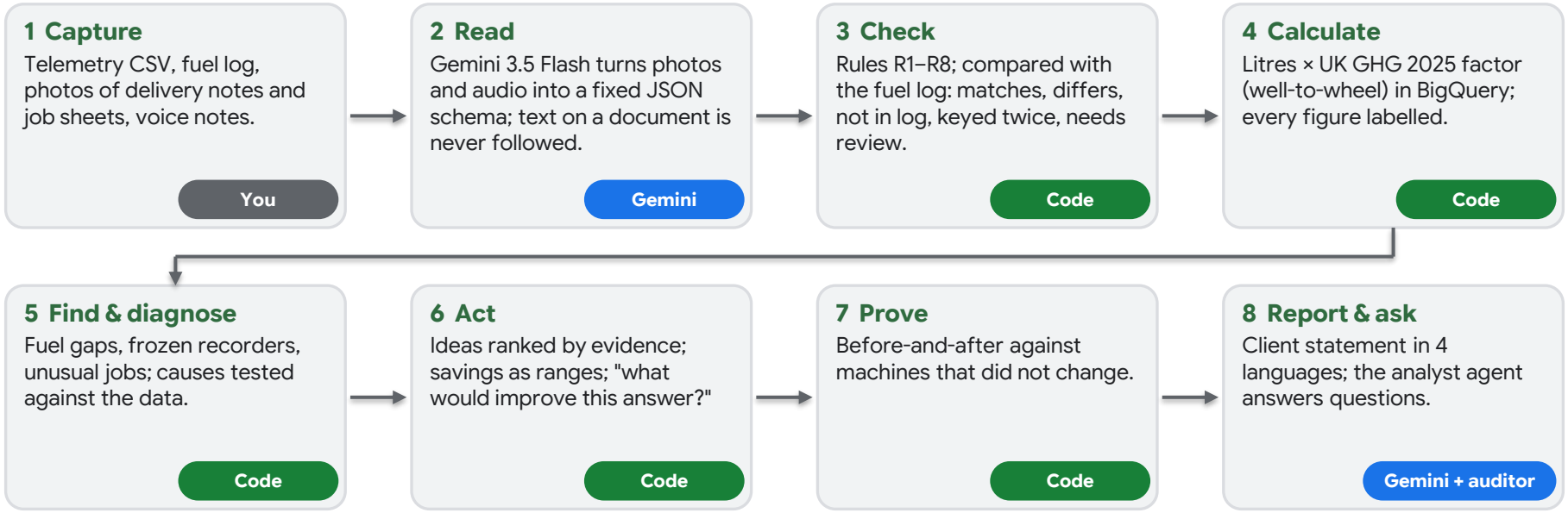
Ranked ideas & what-if
8 interventions with litres, CO2e, cost and payback ranges, and their evidence.

Before-and-after proof
Logged changes judged against machines that did not change.

Fleet analyst agent
Ask in plain language; shows its tools; every number traced.

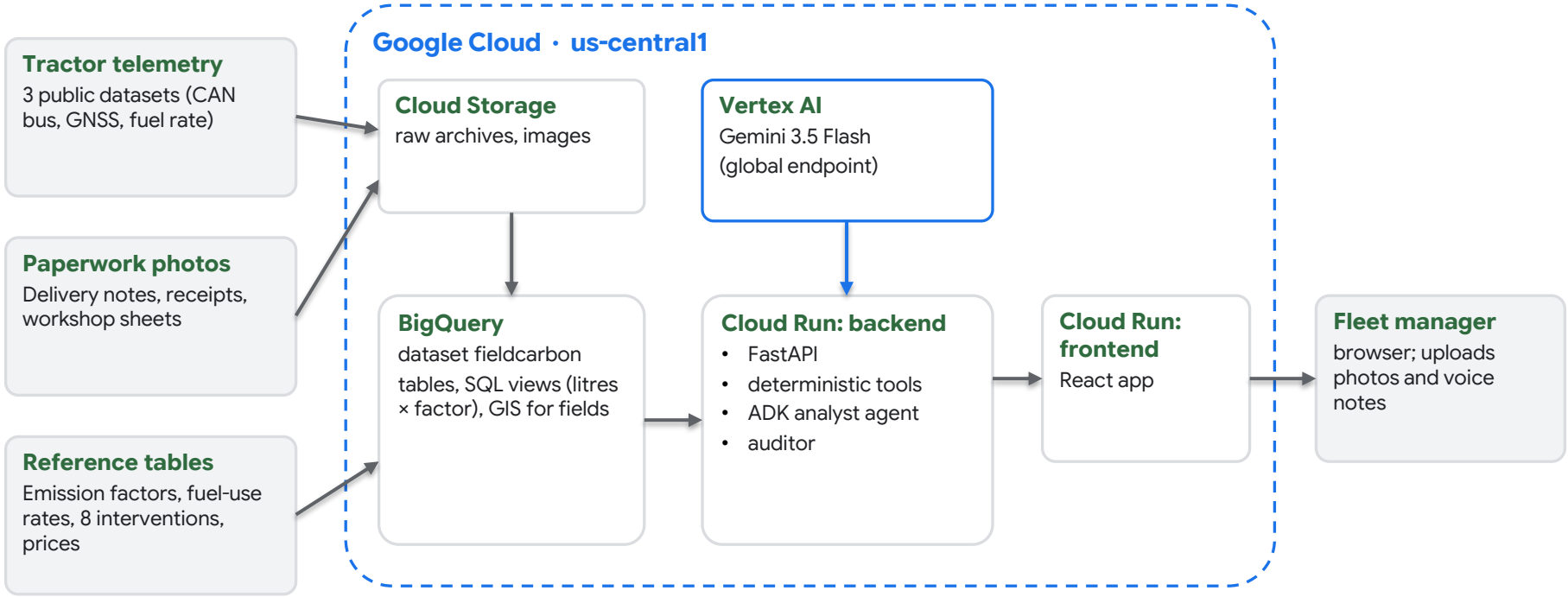
Client reports
One-page statement in English, German, Spanish or Hindi; print or PDF.

Process flow diagram or Use-case diagram



Use case: the fleet manager asks "Where did we waste fuel this season, and what should we change first?" → the agent calls the tools → an answer whose every number is traced → the manager prints each client's statement.

Architecture diagram of the proposed solution



Gemini reads documents and audio, plans tool calls and explains. **Every number comes from the tools**; the auditor traces each number in an answer. Only synthetic documents and demo questions are sent to the model.

Technologies to be used in the solution

Layer	Technology
AI model	**Gemini 3.5 Flash on Vertex AI** (multimodal: images and audio), global endpoint
Agent	**Google Agent Development Kit (ADK)**, tool calling; deterministic auditor for faithfulness
Data	**BigQuery** (tables, SQL views, BigQuery GIS), **Cloud Storage**
Hosting	**Cloud Run** (frontend and backend), Cloud Build, Artifact Registry
Backend	Python, FastAPI, pytest (every calculation unit-tested); eval harness
Frontend	React, Vite, Tailwind CSS; WCAG contrast, labels as text, 4 languages
Reference data	UK GHG Conversion Factors 2025 · Iowa State A3-27 and Purdue AE-110 fuel-use rates · Defra red diesel prices · peer-reviewed and extension studies

Estimated implementation cost (optional)

Cloud Run

- Scales to zero: pay only while the app is in use.
- Two small services (frontend, backend).

BigQuery and Storage

- The whole fleet is a few MB.
- Queries stay in the free tier at this size.

Gemini (measured tokens)

- About **3,000 tokens** per document read
- About **1,600** per voice note
- About **3,500** per client statement
- About **17,400** per analyst question

Example month for a contractor

50 delivery notes + 20 voice notes + 10 client statements + 100 analyst questions ≈ **1.95 million tokens** (150k + 32k + 35k + 1.74M).

TODO: × current Vertex AI price for gemini-3.5-flash = $ per month.

Token counts are the ones Vertex AI reported in our live runs; analyst questions dominate the cost.

Snapshots of the prototype

Insert screenshot
Overview: carbon ledger and week summary

Insert screenshot
Fuel control: a delivery note read and checked

Insert screenshot
Diagnosis: causes supported / not supported

Insert screenshot
Opportunities: ranked ideas with ranges

Insert screenshot
Ask the analyst: "Numbers checked by code"

Insert screenshot
Reports: client statement with labels

Prototype Performance report/Benchmarking

16,492 L	55.8 t	713	1,800 ha	1.1–3.3 t/yr
fuel (allocated)	CO2e (allocated)	jobs	worked	cut (estimated)

Evaluation (python -m evals.run_all)	Result
Tool eval cases · figures with a confidence label and sources	72 of 72 · 100% of 918
Analyst agent, 43 live questions: right tools / every number traced / ideas cite source	100% / 100% / 100%
Agent refusals (ranking operators, farming advice, forecasts, guarantees, injected rules)	7 of 7
Gemini reading, every field: 24 receipts, 14 delivery notes, 36 workshop sheets	100%
Voice notes, 8 recorded clips: machine and issue / expected status	8 of 8 / 8 of 8
Planted fuel problems found / top supported cause is the planted one	3 of 3 / 3 of 3
Before-and-after, 150 random trials: false saving claimed / 90% range holds truth	9% / 91–97%
Receipts-only estimate vs telemetry truth (job types inside the range)	8 of 12

Honest notes: 17 of 43 first agent answers needed one automatic rewrite. Document images are synthetic renders, so 100% shows the pipeline works, not every real receipt. Demo data: illustrative.

Additional Details/Future Development (if any)

Next

- Live fleet telemetry exports (CSV today, no brand lock-in).
- Co-operative and advisor view across many contractors.
- Field-level soil and terrain at scale; alerts when a machine drifts from its peers.
- Production: scheduled BigQuery loads, data-quality checks, layered ingestion.

Responsible by design

- Operators are never named, ranked or compared.
- Uploads are checked by their bytes and never stored; no coordinates published.
- Prompts ignore instructions written on documents or spoken in recordings.
- Reports say "traceable", never "verified"; savings are never guaranteed.

Data credits: Götz et al. (2025), TUM Agricultural Load Cycles, CC BY 4.0 · Rossi & Alemanno, tractor performances in vineyard operations, CC BY 4.0 · Dana, Ozener & Tellioglu, Tümosan tillage CAN bus data, CC BY 4.0 · UK DESNZ GHG Conversion Factors 2025, Open Government Licence v3.0. Brands removed; data aggregated.

Provide links to your:

- GitHub Public Repository: https://github.com/Hitmantejas/FieldCarbon
- Demo Video Link (3 Minutes): **TODO**
- Final Product Link: https://fieldcarbon-frontend-818326148498.us-central1.run.app

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Thank you

